

# Handling nuisance parameters via quasi-reverse information projection

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April 4, 2023

## 1 Universal inference and quasi-reverse information projection

## 2 Asymptotics under regular statistical models

In this section, we analyze the asymptotic performance of quasi-RIPr under regular statistical models. Suppose the data generating distribution is  $p_{\theta^*}$  with  $\theta^* = (\psi^*, \lambda^*)^\top$  where  $\psi^*$  denotes the parameter of interest and  $\lambda^*$  the nuisance parameter. We will write  $p_{\theta^*}$  as  $p_{\psi^*, \lambda^*}$  sometimes. Let  $\hat{\theta}_1 = (\hat{\psi}_1, \hat{\lambda}_1)$  be any estimate of  $\theta^*$  using the first fold of data and  $(Y_1, \dots, Y_{n_0})$  denotes the zeroth fold of data. The quasi-RIPr of  $p_{\psi, \lambda}$  is defined as  $p_{\psi, \hat{\psi}}$  where

$$\hat{\lambda}(\psi) = \arg \min_{\lambda \in \Lambda} \text{KL}(p_{\hat{\theta}_1}, p_{\psi, \lambda}) = \arg \max_{\lambda \in \Lambda} \mathbb{E}_{Y \sim p_{\hat{\theta}_1}} [\log p_{\psi, \lambda}(Y)]. \quad (1)$$

Note that  $\hat{\lambda}(\cdot)$  only depends on the first fold and hence independent of the zeroth fold. Similarly, let

$$\bar{\lambda}(\psi) = \arg \min_{\lambda \in \Lambda} \text{KL}(p_{\theta^*}, p_{\psi, \lambda}) = \arg \max_{\lambda \in \Lambda} \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\psi, \lambda}(Y)]. \quad (2)$$

The assumption stated in the next subsection would guarantee the existence and uniqueness of  $\hat{\lambda}(\psi)$  and  $\bar{\lambda}(\psi)$ . Note that  $\bar{\lambda}(\cdot)$  depends on  $\lambda^*$  which is unknown to the analyst. Furthermore,

$$\hat{\lambda}(\hat{\psi}_1) = \hat{\lambda}_1, \quad \bar{\lambda}(\psi^*) = \lambda^* \quad (3)$$

Then we define an oracle split LRT confidence set for  $\psi$  as

$$\mathcal{C}_{\psi, 1-\alpha}^{\text{Oracle}} = \left\{ \psi : 2 \sum_{i=1}^{n_0} \{\log q_{\hat{\psi}_1}(Y_i) - \log q_{\psi}(Y_i)\} < -2 \log \alpha \right\}. \quad (4)$$

This is analogous to the generalized likelihood ratio test, though the profile likelihood function depends on the unknown true parameter  $\lambda^*$ . We prove that  $\mathcal{C}_{\psi, 1-\alpha}^{\text{RIPr}}$  and  $\mathcal{C}_{\psi, 1-\alpha}^{\text{Oracle}}$  have asymptotically identical type-I error for regular statistical models under the following assumptions.

**A1**  $\log p_{\psi^*, \lambda}(Y)$  is concave in  $\lambda$  on  $\Lambda$ .

**A2** For any  $\psi \in \Psi$ ,  $\mathbb{E}_{\theta^*}[\log p_{\psi, \lambda}(Y)]$  and  $\mathbb{E}_{\hat{\theta}_1}[\log p_{\psi, \lambda}(Y)]$  are differentiable in  $\lambda$  and there exist unique  $\bar{\lambda}(\psi), \hat{\lambda}(\psi) \in \Lambda$  such that

$$\bar{\lambda}(\psi) = \arg \max_{\lambda \in \Lambda} \mathbb{E}_{\theta^*}[\log p_{\psi, \lambda}(Y)], \quad \hat{\lambda}(\psi) = \arg \max_{\lambda \in \Lambda} \mathbb{E}_{\hat{\theta}_1}[\log p_{\psi, \lambda}(Y)].$$

**A3** There exists a constant  $C > 0$  and a neighborhood of  $\theta^*$  of a constant size  $c > 0$  such that for any  $\theta$  in the neighborhood,  $\sqrt{\mathbb{E}_{\theta^*}[(1 - p_{\theta}(Y)/p_{\theta^*}(Y))^2]} \leq C \|\theta - \theta^*\|$ .

**A4** For any  $\theta \in \Psi \times \Lambda$ ,  $\nabla_\theta \log p_\theta(Y)$ ,  $\nabla_{\theta\theta} \log p_\theta(Y)$  are  $\ell(Y)$ -Lipschitz under their corresponding operator norms where  $\sqrt{\mathbb{E}_{\theta^*}[\ell^2(Y)]} \leq L$  for some constant  $L \in (0, \infty)$ .

**A5** There exists  $B \in (0, \infty)$  such that

$$\sqrt{\mathbb{E}_{\theta^*} \|\nabla_\lambda \log p_{\theta^*}(Y)\|^2} \leq B$$

**A6** There exists  $b \in (0, \infty)$  such that

$$\mathbb{E}_{\theta^*} [-\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)] \succeq bI.$$

**Remark 1.** Under assumptions **A1** and **A6**, the uniqueness of  $\bar{\lambda}(\psi')$  and  $\hat{\lambda}(\psi')$  is guaranteed in a neighborhood of  $\psi^*$ , which is sufficient for all following lemmas.

**Theorem 1.** Assume that

$$\|\hat{\theta}_1 - \theta^*\| = o_{\mathbb{P}}(1), \quad \sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\| = o_{\mathbb{P}}(1), \quad n_0 \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|^2 = o_{\mathbb{P}}(1) \quad (5)$$

Under assumptions **A1** - **A6**,

$$2 \sum_{i=1}^{n_0} \{\log p_{\hat{\theta}_1}(Y_i) - \log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y_i)\} = 2 \sum_{i=1}^{n_0} \{\log q_{\hat{\psi}_1}(Y_i) - \log q_{\psi^*}(Y_i)\} + o_{\mathbb{P}}(1).$$

Applying the theoretical results from the Appendix of Wasserman et al. [2020], Theorem 1 implies that the coverage rate only depends on  $k$ , the dimension of the target parameter, instead of  $k + d$ . Thus, in contrast to the original split LRT,  $\mathcal{C}_{\psi, 1-\alpha}^{\text{RIPR}}$  breaks the curse of dimensionality of the nuisance parameter for regular statistical models.

Before moving to the proof of Theorem 1, we show that (4) enjoys certain optimality to the first order. In fact, for any function  $f(\cdot) : \Psi \mapsto \Lambda$ , we can consider a split LRT confidence set as

$$\mathcal{C}_{\psi, 1-\alpha}^f = \left\{ \psi : 2 \sum_{i=1}^{n_0} \{\log p_{\hat{\psi}_1, f(\hat{\psi}_1)}(Y_i) - \log p_{\psi, f(\psi)}(Y_i)\} < -2 \log \alpha \right\}.$$

The following result shows that  $\bar{\lambda}(\cdot)$  maximizes the expectation of the LHS, conditional on  $\hat{\theta}_1$ , at  $\psi = \psi^*$ .

**Theorem 2.**

$$\bar{\lambda} = \arg \max_{f: \Psi \mapsto \Lambda} \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\hat{\psi}_1, f(\hat{\psi}_1)}(Y)] - \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\psi^*, f(\psi^*)}(Y)]$$

*Proof of Theorem 2.* Fix any  $f$ . By (2),

$$\text{KL}(p_{\theta^*}, p_{\hat{\psi}_1, \bar{\lambda}(\hat{\psi}_1)}) \leq \text{KL}(p_{\theta^*}, p_{\hat{\psi}_1, f(\hat{\psi}_1)}) \implies \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\hat{\psi}_1, f(\hat{\psi}_1)}(Y)] \geq \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\hat{\psi}_1, \bar{\lambda}(\hat{\psi}_1)}(Y)].$$

On the other hand,

$$\begin{aligned} 0 &= \text{KL}(p_{\theta^*}, p_{\theta^*}) \leq \text{KL}(p_{\theta^*}, p_{\psi^*, f(\psi^*)}) \\ &\implies \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\psi^*, f(\psi^*)}(Y)] \leq \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\theta^*}(Y)] = \mathbb{E}_{Y \sim p_{\theta^*}} [\log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y)]. \end{aligned}$$

The proof is completed by combining the above two pieces.  $\square$

## 2.1 Proof of Theorem 1

For any  $a \in \mathbb{R}^k, b \in \mathbb{R}^d$ , let  $\nabla_\psi \log p_{a,b}$  (resp.  $\nabla_\lambda \log p_{a,b}$ ) denote the partial derivative of  $\log p_{\psi,\lambda}$  with respect to  $\psi$  (resp.  $\lambda$ ) evaluated at  $a$  and  $b$ , arranged into a vector in  $\mathbb{R}^k$  (resp.  $\mathbb{R}^d$ ). Similarly, let  $\nabla_{\psi\psi} \log p_{a,b} \in \mathbb{R}^{k \times k}$ ,  $\nabla_{\psi\lambda} \log p_{a,b} = (\nabla_\lambda \log p_{a,b})^\top \in \mathbb{R}^{k \times d}$ , and  $\nabla_{\lambda\lambda} \log p_{a,b} \in \mathbb{R}^{d \times d}$  denote the second derivatives. Finally, let  $\nabla_\psi \lambda \in \mathbb{R}^{d \times k}$  the first derivative of  $\lambda \in \{\bar{\lambda}, \hat{\lambda}\}$  with respect to  $\psi$ . To simplify the notation, we rewrite  $\mathbb{E}_{Y \sim p_\theta}$  (resp.  $\mathbb{E}_{Y \sim p_{\psi,\lambda}}$ ) as  $\mathbb{E}_\theta$  (resp.  $\mathbb{E}_{\psi,\lambda}$ ).

First, we state some properties of  $\bar{\lambda}$  and  $\hat{\lambda}$ . The proof is relegated to the end of this section

**Lemma 1.** *Under assumptions A1 - A6, there exist constants  $a, A \in (0, \infty)$  that only depends on  $C, L, B, b$  such that,*

(a) *For any  $\psi' \in \Psi$ ,*

$$\mathbb{E}_{\theta^*} [\nabla_\lambda \log p_{\psi', \bar{\lambda}(\psi')}(Y)] = \mathbb{E}_{\hat{\theta}_1} [\nabla_\lambda \log p_{\psi', \hat{\lambda}(\psi')}(Y)] = 0.$$

(b)  *$\bar{\lambda}(\psi)$  is differentiable at  $\psi^*$  and*

$$\nabla_\psi \bar{\lambda}(\psi^*)^\top \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)] = -\mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)].$$

(c) *For any  $\psi'$  with  $\|\psi' - \psi^*\| < a$ ,*

$$\max \left\{ \|\nabla_\psi \bar{\lambda}(\psi')\|, \|\nabla_\psi \hat{\lambda}(\psi')\| \right\} \leq A.$$

(d) *Assume that  $\|\hat{\theta}_1 - \theta^*\| < a$ . For any  $\psi', \psi''$  with  $\|\psi' - \psi^*\| < a, \|\psi'' - \psi^*\| < a$ ,*

$$\|\nabla_\psi \bar{\lambda}(\psi') - \nabla_\psi \bar{\lambda}(\psi'')\| \leq A\|\psi' - \psi''\|, \quad \|\nabla_\psi \hat{\lambda}(\psi') - \nabla_\psi \hat{\lambda}(\psi'')\| \leq A\|\psi' - \psi''\|.$$

(e) *Assume that  $\|\hat{\theta}_1 - \theta^*\| < a$ . Then*

$$\max \left\{ \|\hat{\lambda}(\psi^*) - \lambda^*\|, \|\nabla_\psi \hat{\lambda}(\psi^*) - \nabla_\psi \bar{\lambda}(\psi^*)\| \right\} \leq A\|\hat{\theta}_1 - \theta^*\|.$$

**Proof of Theorem 1.** Throughout the proof, we will repeatedly use the following fact:

$$\|\hat{\psi} - \psi^*\| \leq \|\hat{\theta}_1 - \theta^*\| = o_P(1).$$

We will restrict ourselves into the event  $\|\hat{\theta}_1 - \theta^*\| < a$  so that Lemma 1 can be applied. Another fact that will be repeatedly invoked is that assumption A4 implies

$$\|\nabla_{\theta\theta} \log p_\theta(Y)\| \leq \ell(Y) \implies \mathbb{E} \|\nabla_{\theta\theta} \log p_\theta(Y)\| \leq L. \quad (6)$$

Recall that  $\hat{\lambda}_1 = \hat{\lambda}(\hat{\psi}_1)$ ,

$$2 \sum_{i=1}^{n_0} \{\log p_{\hat{\theta}_1}(Y_i) - \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i)\} = 2 \sum_{i=1}^{n_0} \{\log p_{\hat{\psi}_1, \hat{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i)\}.$$

By Taylor expansion and assumption A4,

$$\begin{aligned} & \left| 2 \left\{ \log p_{\hat{\psi}_1, \hat{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\} \right. \\ & \quad - 2 \nabla_\psi \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i)^\top (\hat{\psi}_1 - \psi^*) - 2 \nabla_\lambda \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i)^\top (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) \\ & \quad \left. - (\hat{\psi}_1 - \psi^*)^\top \nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) (\hat{\psi}_1 - \psi^*) \right| \end{aligned}$$

$$\begin{aligned}
& - \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right)^\top \nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \\
& - 2 \left( \hat{\psi}_1 - \psi^* \right)^\top \nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \Big| \\
& \leq \ell(Y_i) \left( \|\hat{\psi}_1 - \psi^*\|^3 + \|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\|^3 \right).
\end{aligned} \tag{7}$$

Similarly, recalling that  $\lambda^* = \bar{\lambda}(\psi^*)$ ,

$$2 \sum_{i=1}^{n_0} \{ \log q_{\hat{\psi}_1}(Y_i) - \log q_{\psi^*}(Y_i) \} = 2 \sum_{i=1}^{n_0} \{ \log p_{\hat{\psi}_1, \bar{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y_i) \}.$$

By Taylor expansion and assumption **A4**,

$$\begin{aligned}
& \left| 2 \left\{ \log p_{\hat{\psi}_1, \bar{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\theta^*}(Y_i) \right\} \right. \\
& \quad - 2 \nabla_{\psi} \log p_{\theta^*}(Y_i)^\top (\hat{\psi}_1 - \psi^*) - 2 \nabla_{\lambda} \log p_{\theta^*}(Y_i)^\top (\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)) \\
& \quad - \left( \hat{\psi}_1 - \psi^* \right)^\top \nabla_{\psi\psi} \log p_{\theta^*}(Y_i) (\hat{\psi}_1 - \psi^*) \\
& \quad - \left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right)^\top \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) (\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)) \\
& \quad \left. - 2 \left( \hat{\psi}_1 - \psi^* \right)^\top \nabla_{\psi\lambda} \log p_{\theta^*}(Y_i) (\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)) \right| \\
& \leq \ell(Y_i) \left( \|\hat{\psi}_1 - \psi^*\|^3 + \|\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)\|^3 \right).
\end{aligned} \tag{8}$$

We decompose the remaining proof into seven steps.

(i) By Lemma 1 (d) and (e),

$$\begin{aligned}
& \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) = \nabla_{\psi} \hat{\lambda}(\psi^*)(\hat{\psi}_1 - \psi^*) + O_{\mathbb{P}}(\|\hat{\psi}_1 - \psi^*\|^2) \\
& = \nabla_{\psi} \bar{\lambda}(\psi^*)(\hat{\psi}_1 - \psi^*) + O_{\mathbb{P}}(\|\hat{\psi}_1 - \psi^*\|^2 + \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|) \\
& = \nabla_{\psi} \bar{\lambda}(\psi^*)(\hat{\psi}_1 - \psi^*) + O_{\mathbb{P}}(\|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|),
\end{aligned} \tag{9}$$

and

$$\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) = \nabla_{\psi} \bar{\lambda}(\psi^*)(\hat{\psi}_1 - \psi^*) + O_{\mathbb{P}}(\|\hat{\psi}_1 - \psi^*\|^2), \tag{10}$$

(ii) By assumption **A4**,

$$\left| \nabla_{\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi} \log p_{\theta^*}(Y_i) - \nabla_{\psi\lambda} \log p_{\theta^*}(Y_i) (\hat{\lambda}(\psi^*) - \lambda^*) \right| \leq \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|^2.$$

By Lemma 1 (e) and assumption **A4**,

$$\sum_{i=1}^{n_0} \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|^2 \|\hat{\psi}_1 - \psi^*\| = O_{\mathbb{P}}(n_0 \|\hat{\theta}_1 - \theta^*\|^2 \|\hat{\psi}_1 - \psi^*\|) = o_{\mathbb{P}}(1).$$

Thus,

$$\begin{aligned}
& 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi} \log p_{\theta^*}(Y_i) \right\}^\top (\hat{\psi}_1 - \psi^*) \\
& = 2 (\hat{\psi}_1 - \psi^*)^\top \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\lambda} \log p_{\theta^*}(Y_i) \right\} (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(1).
\end{aligned} \tag{11}$$

By (??) and Chebyshev's inequality,

$$\sum_{i=1}^{n_0} \nabla_{\psi\lambda} \log p_{\theta^*}(Y_i) = n_0 \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] + O_{\mathbb{P}}(\sqrt{n_0}).$$

Therefore,

$$\begin{aligned} & 2 \left\{ \sum_{i=1}^n \nabla_{\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi} \log p_{\theta^*}(Y_i) \right\}^{\top} (\hat{\psi}_1 - \psi^*) \\ &= 2n_0(\hat{\psi}_1 - \psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\lambda}(\psi^*) - \lambda^*\|) \\ &= 2n_0(\hat{\psi}_1 - \psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|) \\ &= 2n_0(\hat{\psi}_1 - \psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(1), \end{aligned} \quad (12)$$

where the second last line applies Lemma 1 (e).

(iii) By assumption **A4**,

$$\left| \nabla_{\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\lambda} \log p_{\theta^*}(Y_i) - \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) (\hat{\lambda}(\psi^*) - \lambda^*) \right| \leq \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|^2.$$

By Lemma 1,

$$\|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\| = O(\|\hat{\psi}_1 - \psi^*\|),$$

and

$$\sum_{i=1}^{n_0} \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|^2 \|\hat{\psi}_1 - \psi^*\| = O_{\mathbb{P}}(n_0 \|\hat{\theta}_1 - \theta^*\|^2 \|\hat{\psi}_1 - \psi^*\|) = o_{\mathbb{P}}(1).$$

Thus,

$$\begin{aligned} & \left| 2 \sum_{i=1}^{n_0} \left\{ \nabla_{\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\lambda} \log p_{\theta^*}(Y_i) - \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) (\hat{\lambda}(\psi^*) - \lambda^*) \right\}^{\top} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) \right| \\ &= o_{\mathbb{P}}(1). \end{aligned} \quad (13)$$

Note that  $\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\theta^*}(Y)] = 0$ . By (9), (10), Chebyshev's inequality, and (6),

$$\begin{aligned} & 2 \sum_{i=1}^{n_0} \left\{ \nabla_{\lambda} \log p_{\theta^*}(Y_i) + \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) (\hat{\lambda}(\psi^*) - \lambda^*) \right\}^{\top} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) - \nabla_{\psi} \bar{\lambda}(\psi^*) (\hat{\psi}_1 - \psi^*)) \\ &= O_{\mathbb{P}} \left( \left\{ \sqrt{n_0} + n_0 \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) \right\} \cdot \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\| \right) \\ &= O_{\mathbb{P}} \left( \sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\| + n_0 \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|^2 \right) \\ &= o_{\mathbb{P}}(1). \end{aligned} \quad (14)$$

Combining (13) and (14), we have that

$$\begin{aligned} & 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\}^{\top} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) \\ &= 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\theta^*}(Y_i) + \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) (\hat{\lambda}(\psi^*) - \lambda^*) \right\}^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*) (\hat{\psi}_1 - \psi^*) + o_{\mathbb{P}}(1). \end{aligned} \quad (15)$$

Similarly,

$$2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\theta^*}(Y_i) \right\}^{\top} (\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*))$$

$$= 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\theta^*}(Y_i) \right\}^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)(\hat{\psi}_1 - \psi^*) + o_{\mathbb{P}}(1). \quad (16)$$

Subtracting (16) from (15), we have

$$\begin{aligned} & 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\}^{\top} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) - 2 \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda} \log p_{\theta^*}(Y_i) \right\}^{\top} (\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)) \\ &= 2(\hat{\psi}_1 - \psi^*)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\lambda\lambda} \log p_{\theta^*}(Y_i) \right\} (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(1) \\ &= 2n_0(\hat{\psi}_1 - \psi^*)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) + O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|) \\ &= 2n_0(\hat{\psi}_1 - \psi^*)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(1), \end{aligned} \quad (17)$$

where the second last line applies (6), Cherbyshev's inequality, and Lemma 1 (c).

(iv) By assumption **A4**,

$$\|\nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi\psi} \log p_{\theta^*}(Y_i)\| \leq \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|.$$

Thus,

$$\begin{aligned} & \left| (\hat{\psi}_1 - \psi^*)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi\psi} \log p_{\theta^*}(Y_i) \right\} (\hat{\psi}_1 - \psi^*) \right| \\ & \leq \sum_{i=1}^{n_0} \ell(Y_i) \|\hat{\psi}_1 - \psi^*\|^2 \|\hat{\lambda}(\psi^*) - \lambda^*\| \\ & = O_{\mathbb{P}}(n_0 \|\hat{\psi}_1 - \psi^*\|^2 \|\hat{\lambda}(\psi^*) - \lambda^*\|) \\ & = O_{\mathbb{P}}(n_0 \|\hat{\psi}_1 - \psi^*\|^2 \|\hat{\theta}_1 - \theta^*\|) \\ & = o_{\mathbb{P}}(1), \end{aligned} \quad (18)$$

where the second last line applies Lemma 1 (e).

(v) By assumption **A4**,

$$\|\nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi\psi} \log p_{\theta^*}(Y_i)\| \leq \ell(Y_i) \|\hat{\lambda}(\psi^*) - \lambda^*\|.$$

Thus,

$$\begin{aligned} & \left| (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*))^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) - \nabla_{\psi\psi} \log p_{\theta^*}(Y_i) \right\} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) \right| \\ & \leq \sum_{i=1}^{n_0} \ell(Y_i) \|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\|^2 \|\hat{\lambda}(\psi^*) - \lambda^*\| \\ & = O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\|^2 \|\hat{\lambda}(\psi^*) - \lambda^*\|) \\ & = O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\|^2 \|\hat{\theta}_1 - \theta^*\|) \\ & = o_{\mathbb{P}}(1), \end{aligned} \quad (19)$$

where the second last line applies Lemma 1 (c). By Chebyshev's inequality and Lemma 1 (c),

$$\left| (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*))^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\theta^*}(Y_i) - n_0 \mathbb{E}_{\theta^*} [\nabla_{\psi\psi} \log p_{\theta^*}(Y)] \right\} (\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)) \right|$$

$$\begin{aligned}
&= O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\|^2) = O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\|^2) \\
&= O_{\mathbb{P}}(\sqrt{n_0} \|\hat{\psi}_1 - \psi^*\| \|\hat{\theta}_1 - \theta^*\|) = o_{\mathbb{P}}(1).
\end{aligned} \tag{20}$$

By (9) and (6),

$$\begin{aligned}
&\left| n_0 \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\psi} \log p_{\theta^*}(Y)] \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \right. \\
&\quad \left. - n_0 \left( \hat{\psi}_1 - \psi^* \right)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\psi} \log p_{\theta^*}(Y)] \nabla_{\psi} \bar{\lambda}(\psi^*) \left( \hat{\psi}_1 - \psi^* \right) \right| \\
&= O_{\mathbb{P}} \left( n_0 \|\hat{\psi}_1 - \psi^*\|^2 \|\hat{\theta}_1 - \theta^*\| \right) \\
&= o_{\mathbb{P}}(1).
\end{aligned} \tag{21}$$

Combining (19) - (21), we obtain that

$$\begin{aligned}
&\left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\} \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \\
&= n_0 \left( \hat{\psi}_1 - \psi^* \right)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\psi} \log p_{\theta^*}(Y)] \nabla_{\psi} \bar{\lambda}(\psi^*) \left( \hat{\psi}_1 - \psi^* \right) + o_{\mathbb{P}}(1).
\end{aligned} \tag{22}$$

Similarly we can prove that

$$\begin{aligned}
&\left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y_i) \right\} \left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right) \\
&= n_0 \left( \hat{\psi}_1 - \psi^* \right)^{\top} \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\psi} \log p_{\theta^*}(Y)] \nabla_{\psi} \bar{\lambda}(\psi^*) \left( \hat{\psi}_1 - \psi^* \right) + o_{\mathbb{P}}(1).
\end{aligned} \tag{23}$$

Subtracting (23) from (22), we obtain that

$$\begin{aligned}
&\left| \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\} \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \right. \\
&\quad \left. - \left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\psi} \log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y_i) \right\} \left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right) \right| = o_{\mathbb{P}}(1).
\end{aligned} \tag{24}$$

(vi) Using the same argument as step (iv) and (v), we can prove that

$$\begin{aligned}
&\left| 2 \left( \hat{\psi}_1 - \psi^* \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\} \left( \hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*) \right) \right. \\
&\quad \left. - 2 \left( \hat{\psi}_1 - \psi^* \right)^{\top} \left\{ \sum_{i=1}^{n_0} \nabla_{\psi\lambda} \log p_{\psi^*, \bar{\lambda}(\psi^*)}(Y_i) \right\} \left( \bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*) \right) \right| = o_{\mathbb{P}}(1).
\end{aligned} \tag{25}$$

(vii) By (9), (10), and assumption **A4**,

$$\begin{aligned}
&\sum_{i=1}^{n_0} \ell(Y_i) \left( \|\hat{\psi}_1 - \psi^*\|^3 + \|\hat{\lambda}(\hat{\psi}_1) - \hat{\lambda}(\psi^*)\|^3 + \|\bar{\lambda}(\hat{\psi}_1) - \bar{\lambda}(\psi^*)\|^3 \right) \\
&= O_{\mathbb{P}}(n_0 \|\hat{\psi}_1 - \psi^*\|^3) = o_{\mathbb{P}}(1).
\end{aligned} \tag{26}$$

Putting (7), (8), (12), (17), (18), (24), (25), and (26) together, we conclude that

$$\begin{aligned}
&2 \left\{ \log p_{\hat{\psi}_1, \hat{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y_i) \right\} \\
&= 2 \left\{ \log p_{\hat{\psi}_1, \bar{\lambda}(\hat{\psi}_1)}(Y_i) - \log p_{\theta^*}(Y_i) \right\} \\
&\quad + 2n_0 (\hat{\psi}_1 - \psi^*)^{\top} \left\{ \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] + \nabla_{\psi} \bar{\lambda}(\psi^*)^{\top} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)] \right\} (\hat{\lambda}(\psi^*) - \lambda^*) + o_{\mathbb{P}}(1).
\end{aligned}$$

The first term in the last line is zero by Lemma 1. The proof is then completed.

## 2.2 Proof of Lemma 1

We decompose the proof into nine lemmas.

**Lemma 2.** *Under assumption A3, for any function  $f$ ,*

$$|\mathbb{E}_{\hat{\theta}_1}[f(Y)] - \mathbb{E}_{\theta^*}[f(Y)]| \leq C\sqrt{\mathbb{E}_{\theta^*}[f^2(Y)]} \cdot \|\hat{\theta}_1 - \theta^*\|.$$

*Proof.* By definition,

$$\mathbb{E}_{\hat{\theta}_1}[f(Y)] - \mathbb{E}_{\theta^*}[f(Y)] = \mathbb{E}_{\theta^*} \left[ f(Y) \left( \frac{p_{\hat{\theta}_1}(Y)}{p_{\theta^*}(Y)} - 1 \right) \right].$$

The lemma follows from Cauchy-Schwarz inequality and assumption A2.  $\square$

**Lemma 3.** *Under assumptions A1 - A5, for any  $\psi' \in \Psi$ ,*

$$\mathbb{E}_{\theta^*}[\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] = \mathbb{E}_{\hat{\theta}_1}[\nabla_{\lambda} \log p_{\psi', \hat{\lambda}(\psi')}(Y)] = 0.$$

*Proof.* By assumption A1 and A2,

$$\nabla_{\lambda} \mathbb{E}_{\theta^*}[\log p_{\psi', \lambda}(Y)]|_{\lambda=\bar{\lambda}(\psi')} = \nabla_{\lambda} \mathbb{E}_{\hat{\theta}_1}[\log p_{\psi', \lambda}(Y)]|_{\lambda=\hat{\lambda}(\psi')} = 0$$

Since  $\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)$  is  $\ell(Y)$ -Lipschitz, for any  $\psi' \in \Psi$ ,

$$\|\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)\| \leq \|\nabla_{\lambda} \log p_{\theta^*}(Y)\| + \eta \ell(Y), \quad \text{where } \eta = \|\psi' - \psi^*\| + \|\bar{\lambda}(\psi') - \lambda^*\|. \quad (27)$$

The RHS is integrable by assumptions A4 and A5. By the rule of differentiation under the integral sign (measure theory version of Leibniz's integral rule),

$$\mathbb{E}_{\theta^*}[\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] = \nabla_{\lambda} \mathbb{E}_{\theta^*}[\log p_{\psi', \lambda}(Y)]|_{\lambda=\bar{\lambda}(\psi')} = 0.$$

Similar to (27),

$$\|\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)\| \leq \|\nabla_{\lambda} \log p_{\theta^*}(Y)\| + \hat{\eta} \ell(Y), \quad \text{where } \hat{\eta} = \|\psi' - \psi^*\| + \|\hat{\lambda}(\psi') - \lambda^*\|.$$

To prove the second claim of the lemma, we just need to prove

$$\mathbb{E}_{\hat{\theta}_1} \|\nabla_{\lambda} \log p_{\theta^*}(Y)\| < \infty, \quad \mathbb{E}_{\hat{\theta}_1}[\ell(Y)] < \infty$$

By Lemma 2,

$$\begin{aligned} \mathbb{E}_{\hat{\theta}_1} \|\nabla_{\lambda} \log p_{\theta^*}(Y)\| &\leq \mathbb{E}_{\theta^*} \|\nabla_{\lambda} \log p_{\theta^*}(Y)\| + C\sqrt{\mathbb{E}_{\theta^*} \|\nabla_{\lambda} \log p_{\theta^*}(Y)\|^2} \cdot \|\hat{\theta}_1 - \theta^*\| \\ &\leq L \left( 1 + C\|\hat{\theta}_1 - \theta^*\| \right) < \infty, \end{aligned}$$

and

$$\mathbb{E}_{\hat{\theta}_1}[\ell(Y)] \leq \mathbb{E}_{\theta^*}[\ell(Y)] + C\sqrt{\mathbb{E}_{\theta^*}[\ell(Y)^2]} \cdot \|\hat{\theta}_1 - \theta^*\| \leq L \left( 1 + C\|\hat{\theta}_1 - \theta^*\| \right) < \infty. \quad (28)$$

The proof is then complete.  $\square$

**Lemma 4.** *Under assumptions A1, A2, A4, and A6, there exist  $a_1, \bar{L}_{\lambda} \in (0, \infty)$  that only depend on  $b$  and  $L$ , such that*

$$\|\psi' - \psi^*\| \leq a_1, \|\psi'' - \psi^*\| \leq a_1 \implies \|\bar{\lambda}(\psi') - \bar{\lambda}(\psi'')\| \leq \bar{L}_{\lambda} \|\psi' - \psi''\|.$$

*Proof.* We first prove the lemma in the case  $\psi'' = \psi^*$ . By Lemma 3 and assumption **A4**,

$$\begin{aligned} & \left\| \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)] \right\| \\ &= \left\| \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] \right\| \\ &\leq \mathbb{E}_{\theta^*} [\ell(Y)] \cdot \|\psi' - \psi^*\| \leq L \|\psi' - \psi^*\|. \end{aligned} \quad (29)$$

On the other hand, by Taylor's expansion and the fact that  $\lambda^* = \bar{\lambda}(\psi^*)$ ,

$$\begin{aligned} & \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)]^{\top} (\bar{\lambda}(\psi') - \lambda^*) \\ &= \left( \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\theta^*}(Y)] \right)^{\top} (\bar{\lambda}(\psi') - \lambda^*) \\ &= \int_0^1 (\bar{\lambda}(\psi') - \lambda^*)^{\top} \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi^*, t\bar{\lambda}(\psi') + (1-t)\lambda^*}(Y) \right] (\bar{\lambda}(\psi') - \lambda^*) dt. \end{aligned} \quad (30)$$

By assumption **A4**,

$$\begin{aligned} \left\| \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi^*, t\bar{\lambda}(\psi') + (1-t)\lambda^*}(Y) \right] - \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi^*, \lambda^*}(Y)] \right\| &\leq \mathbb{E}_{\theta^*} [\ell(Y)] \cdot t \|\bar{\lambda}(\psi') - \lambda^*\| \\ &\leq Lt \|\bar{\lambda}(\psi') - \lambda^*\|. \end{aligned}$$

Together with assumption **A6**, the above inequality implies that, for any  $t \leq b/2L \|\bar{\lambda}(\psi') - \lambda^*\|$ ,

$$\mathbb{E}_{\theta^*} \left[ -\nabla_{\lambda\lambda} \log p_{\psi^*, t\bar{\lambda}(\psi') + (1-t)\lambda^*}(Y) \right] \succeq (b/2)I.$$

Thus,

$$\begin{aligned} & \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)]^{\top} (\bar{\lambda}(\psi') - \lambda^*) \\ &\geq \frac{b}{2} \int_0^1 \|\bar{\lambda}(\psi') - \lambda^*\|^2 I \left( t \leq \frac{b}{2L \|\bar{\lambda}(\psi') - \lambda^*\|} \right) dt \\ &= \frac{b}{2} \|\bar{\lambda}(\psi') - \lambda^*\| \cdot \min \left\{ \|\bar{\lambda}(\psi') - \lambda^*\|, \frac{b}{2L} \right\}. \end{aligned}$$

By Cauchy-Schwarz inequality,

$$|\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)]^{\top} (\bar{\lambda}(\psi') - \lambda^*)| \leq \|\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)]\| \cdot \|\bar{\lambda}(\psi') - \lambda^*\|.$$

Therefore,

$$\|\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi^*, \bar{\lambda}(\psi')}(Y)]\| \geq \frac{b}{2} \min \left\{ \|\bar{\lambda}(\psi') - \lambda^*\|, \frac{b}{2L} \right\}.$$

By (29), we obtain that

$$\min \left\{ \|\bar{\lambda}(\psi') - \lambda^*\|, \frac{b}{2L} \right\} \leq \frac{2L}{b} \|\psi' - \psi^*\|.$$

Thus, when  $\|\psi' - \psi^*\| \leq (b/2L)^2$ ,

$$\|\bar{\lambda}(\psi') - \lambda^*\| \leq \frac{2L}{b} \|\psi' - \psi^*\|.$$

Now we move to the general case. Similar to (29), we have

$$\left\| \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi'', \bar{\lambda}(\psi')}(Y)] \right\| \leq L \|\psi' - \psi''\|. \quad (31)$$

Similar to (30), we obtain that

$$\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi'', \bar{\lambda}(\psi')}(Y)]^{\top} (\bar{\lambda}(\psi') - \bar{\lambda}(\psi''))$$

$$= \int_0^1 (\bar{\lambda}(\psi') - \bar{\lambda}(\psi''))^\top \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\bar{\lambda}(\psi') + (1-t)\bar{\lambda}(\psi'')}(Y) \right] (\bar{\lambda}(\psi') - \bar{\lambda}(\psi'')) dt. \quad (32)$$

By assumption **A4**,

$$\begin{aligned} & \left\| \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\bar{\lambda}(\psi') + (1-t)\bar{\lambda}(\psi'')}(Y) \right] - \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\theta^*}(Y) \right] \right\| \\ & \leq L (\|\psi'' - \psi^*\| + t\|\bar{\lambda}(\psi') - \lambda^*\| + (1-t)\|\bar{\lambda}(\psi'') - \lambda^*\|). \end{aligned}$$

When  $\|\psi' - \psi^*\|, \|\psi'' - \psi^*\| \leq \min\{b^2/8L^2, b/4L\}$ , we proved that

$$\|\bar{\lambda}(\psi') - \lambda^*\|, \|\bar{\lambda}(\psi'') - \lambda^*\| \leq \frac{2L}{b} \cdot \frac{b^2}{8L^2} = \frac{b}{4L}.$$

As a result, for any  $t \in [0, 1]$ ,

$$\left\| \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\bar{\lambda}(\psi') + (1-t)\bar{\lambda}(\psi'')}(Y) \right] - \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\theta^*}(Y) \right] \right\| \leq \frac{b}{2}.$$

Together with assumption **A6**, it implies that, for any  $t \in [0, 1]$ ,

$$\mathbb{E}_{\theta^*} \left[ -\nabla_{\lambda\lambda} \log p_{\psi'', t\bar{\lambda}(\psi') + (1-t)\bar{\lambda}(\psi'')}(Y) \right] \succeq (b/2)I. \quad (33)$$

By (32), we obtain that

$$\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi'', \bar{\lambda}(\psi')}(Y)]^\top (\bar{\lambda}(\psi') - \bar{\lambda}(\psi'')) \geq \frac{b}{2} \|\bar{\lambda}(\psi') - \bar{\lambda}(\psi'')\|^2.$$

By Cauchy-Schwarz inequality and (31),

$$\|\bar{\lambda}(\psi') - \bar{\lambda}(\psi'')\| \leq \frac{2L}{b} \|\psi' - \psi''\|.$$

Therefore, the Lemma holds with  $a_1 = \min\{b^2/8L^2, b/4L\}$  and  $\bar{L}_\lambda = 2L/b$ .  $\square$

**Lemma 5.** *Under assumptions **A1**, **A2**, **A4**, and **A6**, there exist  $\hat{L}_\lambda \in (0, \infty)$  that only depend on  $C, L$ , and  $b$ , such that*

$$\|\hat{\theta}_1 - \theta^*\| \leq a_1, \|\psi' - \psi^*\| \leq a_1, \|\psi'' - \psi^*\| \leq a_1 \implies \|\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')\| \leq \hat{L}_\lambda \|\psi' - \psi''\|,$$

where  $a_1$  is the constant defined in Lemma 4.

*Proof.* By (28), assumption **A4** holds with  $L$  replaced by  $L + C\|\hat{\theta}_1 - \theta^*\|$ . Similar to (31) and (32), we have

$$\left\| \mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda} \log p_{\psi'', \hat{\lambda}(\psi')}(Y)] \right\| \leq (L + C\|\hat{\theta}_1 - \theta^*\|) \|\psi' - \psi''\|, \quad (34)$$

and

$$\begin{aligned} & \mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda} \log p_{\psi'', \hat{\lambda}(\psi')}(Y)]^\top (\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')) \\ & = \int_0^1 (\hat{\lambda}(\psi') - \hat{\lambda}(\psi''))^\top \mathbb{E}_{\hat{\theta}_1} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\hat{\lambda}(\psi') + (1-t)\hat{\lambda}(\psi'')}(Y) \right] (\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')) dt. \end{aligned} \quad (35)$$

By Lemma 2 and assumption **A4**, when  $\|\psi' - \psi^*\|, \|\psi'' - \psi^*\| \leq a_1$ ,

$$\left\| \mathbb{E}_{\hat{\theta}_1} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\hat{\lambda}(\psi') + (1-t)\hat{\lambda}(\psi'')}(Y) \right] - \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', t\hat{\lambda}(\psi') + (1-t)\hat{\lambda}(\psi'')}(Y) \right] \right\| \leq L\|\hat{\theta}_1 - \theta^*\|.$$

By (33), when  $\|\hat{\theta}_1 - \theta^*\| \leq a_1 \leq b/4L$ ,

$$\mathbb{E}_{\hat{\theta}_1} \left[ -\nabla_{\lambda\lambda} \log p_{\psi'', t\bar{\lambda}(\psi') + (1-t)\bar{\lambda}(\psi'')}(Y) \right] \succeq (b/4)I. \quad (36)$$

By (35),

$$\mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda} \log p_{\psi'', \hat{\lambda}(\psi')}(Y)]^{\top} (\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')) \geq \frac{b}{4} \|\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')\|^2.$$

Together with (34), it implies that

$$\|\hat{\lambda}(\psi') - \hat{\lambda}(\psi'')\| \leq \frac{4(L + C\|\hat{\theta}_1 - \theta^*\|)}{b} \leq \frac{4(L + a_1)}{b} \|\psi' - \psi''\|.$$

Therefore, we prove the lemma with  $\hat{L}_{\lambda} = 4(L + a_1)/b$ .  $\square$

**Lemma 6.** Assume that  $\|\hat{\theta}_1 - \theta^*\| < a_1$  where  $a_1$  is the constant defined in Lemma 4. Under assumptions **A1-A6**, for any  $\psi'$  with  $\|\psi' - \psi^*\| < a_1$ ,  $\bar{\lambda}$  and  $\hat{\lambda}$  are differentiable at  $\psi'$  and

$$\begin{aligned} \nabla_{\psi} \bar{\lambda}(\psi')^{\top} \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] &= -\mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)], \\ \nabla_{\psi} \hat{\lambda}(\psi')^{\top} \mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda\lambda} \log p_{\psi', \hat{\lambda}(\psi')}(Y)] &= -\mathbb{E}_{\hat{\theta}_1} [\nabla_{\psi\lambda} \log p_{\psi', \hat{\lambda}(\psi')}(Y)]. \end{aligned}$$

*Proof.* Fix any  $\psi'$  with  $\|\psi' - \psi^*\| < a_1$ . By assumption **A4**, for any sequence  $\psi_n \rightarrow \psi'$  with  $\|\psi_n - \psi'\| \leq a_1 - \|\psi' - \psi^*\|$  for all  $n$  (without loss of generality),

$$\begin{aligned} &\|\nabla_{\lambda} \log p_{\psi_n, \bar{\lambda}(\psi_n)}(Y) - \nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \\ &\quad - \nabla_{\lambda\psi} \log p_{\psi', \bar{\lambda}(\psi')}(Y)(\psi_n - \psi') - \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)(\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi'))\| \\ &\leq \ell(Y) \cdot \|\psi_n - \psi'\|^2 + \|\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi')\|^2. \end{aligned}$$

Since  $\|\psi_n - \psi^*\| \leq \|\psi' - \psi^*\| + \|\psi_n - \psi'\| \leq a_1$ , Lemma 4 implies that

$$\|\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi')\| \leq \bar{L}_{\lambda} \|\psi_n - \psi'\|.$$

Thus,

$$\begin{aligned} &\|\nabla_{\lambda} \log p_{\psi_n, \bar{\lambda}(\psi_n)}(Y) - \nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \\ &\quad - \nabla_{\lambda\psi} \log p_{\psi', \bar{\lambda}(\psi')}(Y)(\psi_n - \psi') - \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)(\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi'))\| \\ &\leq (\ell(Y) + \bar{L}_{\lambda}^2) \|\psi_n - \psi'\|^2. \end{aligned}$$

Note that  $\mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi_n, \bar{\lambda}(\psi_n)}(Y)] = \mathbb{E}_{\theta^*} [\nabla_{\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] = 0$ . Then

$$\begin{aligned} &\|\mathbb{E}_{\theta^*} [\nabla_{\lambda\psi} \log p_{\psi', \bar{\lambda}(\psi')}(Y)](\psi_n - \psi') + \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)](\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi'))\| \\ &\leq \mathbb{E}_{\theta^*} (\ell(Y) + \bar{L}_{\lambda}^2) \|\psi_n - \psi'\|^2 \leq (L + \bar{L}_{\lambda}^2) \|\psi_n - \psi'\|^2. \end{aligned} \tag{37}$$

By (33) with  $\psi'' = \psi'$ ,

$$\mathbb{E}_{\theta^*} [-\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] \succeq (b/2)I \succ 0.$$

Letting  $n \rightarrow \infty$  in (37), we have

$$\bar{\lambda}(\psi_n) - \bar{\lambda}(\psi') = - \left( \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] \right)^{-1} \mathbb{E}_{\theta^*} [\nabla_{\lambda\psi} \log p_{\psi', \bar{\lambda}(\psi')}(Y)](\psi_n - \psi') + o(\|\psi_n - \psi'\|).$$

Since it holds for any sequence  $(\psi_n)$  that converges to  $\psi'$ , we conclude that  $\nabla_{\psi} \bar{\lambda}(\psi')$  exists and

$$\nabla_{\psi} \bar{\lambda}(\psi')^{\top} \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)] = -\mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y)].$$

The results for  $\nabla_{\psi} \hat{\lambda}(\psi')$  can be proved using the same argument, except that Lemma 4 and (33) are replaced by Lemma 5 and (36), respectively.  $\square$

**Lemma 7.** Assume that  $\|\hat{\theta}_1 - \theta^*\| < a_1$  where  $a_1$  is the constant defined in Lemma 4. Under assumptions **A1-A6**, for any  $\psi'$  with  $\|\psi' - \psi^*\| < a_1$ ,

$$\max \left\{ \|\nabla_{\psi} \bar{\lambda}(\psi')\|, \|\nabla_{\psi} \hat{\lambda}(\psi')\| \right\} \leq \bar{L}_{\lambda}.$$

*Proof.* This is a direct consequence of Lemma 4, Lemma 5, and Lemma 6.  $\square$

**Lemma 8.** Under assumptions **A1-A6**, there exist  $\bar{L}_{\nabla\lambda}, \hat{L}_{\nabla\lambda} \in (0, \infty)$  that only depend on  $C, L$ , and  $b$ , such that

$$\|\psi' - \psi^*\| \leq a_1, \|\psi'' - \psi^*\| \leq a_1 \implies \|\nabla_{\psi} \bar{\lambda}(\psi') - \nabla_{\psi} \bar{\lambda}(\psi'')\| \leq \bar{L}_{\nabla\lambda} \|\psi' - \psi''\|,$$

and

$$\|\hat{\theta}_1 - \theta^*\| \leq a_1, \|\psi' - \psi^*\| \leq a_1, \|\psi'' - \psi^*\| \leq a_1 \implies \|\nabla_{\psi} \hat{\lambda}(\psi') - \nabla_{\psi} \hat{\lambda}(\psi'')\| \leq \hat{L}_{\nabla\lambda} \|\psi' - \psi''\|.$$

*Proof.* We first prove the claim for  $\nabla_{\psi} \bar{\lambda}$ . Fix any  $\psi', \psi''$  with  $\|\psi' - \psi^*\|, \|\psi'' - \psi^*\| \leq a_1$ . By (33),

$$\mathbb{E}_{\theta^*} \left[ -\nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \succeq (b/2)I, \quad \mathbb{E}_{\theta^*} \left[ -\nabla_{\lambda\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] \succeq (b/2)I. \quad (38)$$

By assumption **A4**,

$$\mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \leq L, \quad \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] \leq L \quad (39)$$

By Lemma 6,

$$\nabla_{\psi} \bar{\lambda}(\psi') = - \left( \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \right)^{-1} \mathbb{E}_{\theta^*} \left[ \nabla_{\psi\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right],$$

and

$$\nabla_{\psi} \bar{\lambda}(\psi'') = - \left( \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] \right)^{-1} \mathbb{E}_{\theta^*} \left[ \nabla_{\psi\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right].$$

When  $\|\psi' - \psi^*\|, \|\psi'' - \psi^*\| \leq a_1$ , by assumption **A4** and Lemma 4,

$$\begin{aligned} & \left\| \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] - \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \right\| \\ & \leq L (\|\psi' - \psi''\| + \|\nabla_{\psi} \bar{\lambda}(\psi') - \nabla_{\psi} \bar{\lambda}(\psi'')\|) \\ & \leq L(1 + \bar{L}_{\lambda}) \|\psi' - \psi''\|. \end{aligned}$$

Using the identity  $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ , we have that

$$\left\| \left( \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] \right)^{-1} - \left( \mathbb{E}_{\theta^*} \left[ \nabla_{\lambda\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \right)^{-1} \right\| \leq \frac{4L(1 + \bar{L}_{\lambda})}{b^2} \|\psi' - \psi''\|. \quad (40)$$

By assumption **A4** and Lemma 4 again,

$$\begin{aligned} & \left\| \mathbb{E}_{\theta^*} \left[ \nabla_{\psi\lambda} \log p_{\psi'', \bar{\lambda}(\psi'')}(Y) \right] - \mathbb{E}_{\theta^*} \left[ \nabla_{\psi\lambda} \log p_{\psi', \bar{\lambda}(\psi')}(Y) \right] \right\| \\ & \leq L (\|\psi' - \psi''\| + \|\nabla_{\psi} \bar{\lambda}(\psi') - \nabla_{\psi} \bar{\lambda}(\psi'')\|) \\ & \leq L(1 + \bar{L}_{\lambda}) \|\psi' - \psi''\|. \end{aligned} \quad (41)$$

Putting (38) - (41) together, we obtain that

$$\begin{aligned} & \|\nabla_{\psi} \bar{\lambda}(\psi') - \nabla_{\psi} \bar{\lambda}(\psi'')\| \\ & \leq \|\psi' - \psi''\| \left\{ \frac{4L^2(1 + \bar{L}_{\lambda})}{b^2} + \frac{2L(1 + \bar{L}_{\lambda})}{b} + \frac{4L^2(1 + \bar{L}_{\lambda})^2}{b^2} \|\psi' - \psi''\| \right\} \end{aligned}$$

$$\leq \|\psi' - \psi''\| \left\{ \frac{4L^2(1 + \bar{L}_\lambda)}{b^2} + \frac{2L(1 + \bar{L}_\lambda)}{b} + \frac{4a_1L^2(1 + \bar{L}_\lambda)^2}{b^2} \right\}.$$

Thus, the result for  $\nabla_\psi \bar{\lambda}$  holds with  $\bar{L}_{\nabla \lambda} = \frac{4L^2(1 + \bar{L}_\lambda)}{b^2} + \frac{2L(1 + \bar{L}_\lambda)}{b} + \frac{4a_1L^2(1 + \bar{L}_\lambda)^2}{b^2}$ . The result for  $\nabla_\psi \hat{\lambda}$  can be proved via the same argument, except that  $b/2$  is replaced by  $b/4$  in (38) (according to (36)),  $L$  is replaced by  $L + Ca_1$  in (39) (according to (28)), and  $\bar{L}_\lambda$  is replaced by  $\hat{L}_\lambda$  in (40) and (41).  $\square$

**Lemma 9.** *Under assumptions **A1** - **A6**, there exists  $a_2, L' \in (0, \infty)$  that only depend on  $C, L, B$ , and  $b$ , such that*

$$\|\hat{\theta}_1 - \theta^*\| \leq a_2 \implies \max \left\{ \|\hat{\lambda}(\psi^*) - \lambda^*\|, \|\nabla_\psi \hat{\lambda}(\psi^*) - \nabla_\psi \bar{\lambda}(\psi^*)\| \right\} \leq L' \|\hat{\theta}_1 - \theta^*\|.$$

*Proof.* By (36) with  $\psi' = \psi'' = \psi^*$ , when  $\|\hat{\theta}_1 - \theta^*\| \leq a_1$ ,

$$\mathbb{E}_{\hat{\theta}_1}[-\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)] \succeq (b/4)I.$$

By (28), Lemma 2, and assumption **A4**,

$$\left\| \mathbb{E}_{\hat{\theta}_1}[\nabla_{\lambda\lambda} \log p_{\psi^*, \lambda'}(Y)] - \mathbb{E}_{\hat{\theta}_1}[\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)] \right\| \leq (L + Ca_1) \|\lambda' - \lambda^*\|.$$

This implies

$$\|\lambda' - \lambda^*\| \leq \frac{b}{8(L + Ca_1)} \implies \mathbb{E}_{\hat{\theta}_1}[-\nabla_{\lambda\lambda} \log p_{\psi^*, \lambda}(Y)] \succeq (b/8)I. \quad (42)$$

By Lemma 3 and Taylor expansion,

$$\begin{aligned} & -\mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)]^\top (\hat{\lambda}(\psi^*) - \lambda^*) \\ &= \left( \mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] - \mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)] \right)^\top (\hat{\lambda}(\psi^*) - \lambda^*) \\ &= \int_0^1 (\hat{\lambda}(\psi^*) - \lambda^*)^\top \mathbb{E}_{\hat{\theta}_1} \left[ \nabla_{\lambda\lambda} \log p_{\psi^*, t\hat{\lambda}(\psi^*) + (1-t)\lambda^*}(Y) \right] (\hat{\lambda}(\psi^*) - \lambda^*) dt \\ &\geq \frac{b}{8} \int_0^1 \|\hat{\lambda}(\psi^*) - \lambda^*\|^2 I \left( t \leq \frac{b}{8(L + Ca_1)\|\hat{\lambda}(\psi^*) - \lambda^*\|} \right) dt \\ &= \frac{b}{8} \|\hat{\lambda}(\psi^*) - \lambda^*\| \min \left\{ \|\hat{\lambda}(\psi^*) - \lambda^*\|, \frac{b}{8(L + Ca_1)} \right\}, \end{aligned}$$

where the second last inequality follows from assumption **A1** and (42). To bound the LHS, we apply Cauchy Schwarz inequality which implies

$$-\mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)]^\top (\hat{\lambda}(\psi^*) - \lambda^*) \leq \|\mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)]\| \cdot \|\hat{\lambda}(\psi^*) - \lambda^*\|.$$

By Lemma 2, Lemma 3, and assumption **A5**

$$\begin{aligned} \|\mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)]\| &= \|\mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)] - \mathbb{E}_{\hat{\theta}_1}[\nabla_\lambda \log p_{\theta^*}(Y)]\| \\ &\leq \sqrt{\mathbb{E}_{\theta^*} \|\nabla_\lambda \log p_{\theta^*}(Y)\|^2} \cdot \|\hat{\theta}_1 - \theta^*\| \\ &\leq B \|\hat{\theta}_1 - \theta^*\|. \end{aligned}$$

Putting pieces together, we obtain that

$$\min \left\{ \|\hat{\lambda}(\psi^*) - \lambda^*\|, \frac{b}{8(L + Ca_1)} \right\} \leq \frac{8B}{b} \|\hat{\theta}_1 - \theta^*\|.$$

Therefore,

$$\|\hat{\theta}_1 - \theta^*\| \leq \frac{b^2}{64B^2(L + Ca_1)} \implies \|\hat{\lambda}(\psi^*) - \lambda^*\| \leq \frac{8B}{b} \|\hat{\theta}_1 - \theta^*\|. \quad (43)$$

Next, we bound  $\|\nabla_{\psi}\hat{\lambda}(\psi^*) - \nabla_{\psi}\bar{\lambda}(\psi^*)\|$  assuming that  $\|\hat{\theta}_1 - \theta^*\| \leq b^2/64B^2(L + Ca_1)$ . By Lemma 6 with  $\psi' = \psi^*$ ,

$$\nabla_{\psi}\bar{\lambda}(\psi^*) = -(\mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)])^{-1} \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)],$$

and

$$\nabla_{\psi}\hat{\lambda}(\psi^*) = -\left(\mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)]\right)^{-1} \mathbb{E}_{\hat{\theta}_1} [\nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)].$$

By assumption **A4** and Lemma 2,

$$\left\|\mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)]\right\| \leq L\|\hat{\theta}_1 - \theta^*\|,$$

and

$$\left\|\mathbb{E}_{\hat{\theta}_1} [\nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)]\right\| \leq L\|\hat{\theta}_1 - \theta^*\|.$$

By (43) and assumption **A4**,

$$\left\|\mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)]\right\| \leq \frac{8LB}{b}\|\hat{\theta}_1 - \theta^*\|,$$

and

$$\left\|\mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] - \mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)]\right\| \leq \frac{8LB}{b}\|\hat{\theta}_1 - \theta^*\|.$$

Let

$$a_2 = \min \left\{ \frac{b^2}{64B^2(L + Ca_1)}, \frac{b}{2L(1 + 8B/b)} \right\},$$

and assume further that

$$\|\hat{\theta}_1 - \theta^*\| \leq a_2.$$

By assumption **A6** and the above inequalities,

$$\mathbb{E}_{\hat{\theta}_1} [-\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)] \succeq (b/2)I.$$

Using the identity that  $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ ,

$$\left\|\left(\mathbb{E}_{\hat{\theta}_1} [\nabla_{\lambda\lambda} \log p_{\psi^*, \hat{\lambda}(\psi^*)}(Y)]\right)^{-1} - (\mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)])^{-1}\right\| \leq \frac{2L(1 + 8B/b)}{b^2}\|\hat{\theta}_1 - \theta^*\|.$$

Putting pieces together, we obtain that

$$\begin{aligned} & \left\|\nabla_{\psi}\hat{\lambda}(\psi^*) - \nabla_{\psi}\bar{\lambda}(\psi^*)\right\| \\ & \leq \|\hat{\theta}_1 - \theta^*\| \left\{ \frac{8LB}{b} \left\|(\mathbb{E}_{\theta^*} [\nabla_{\lambda\lambda} \log p_{\theta^*}(Y)])^{-1}\right\| + \frac{2L(1 + 8B/b)}{b^2} \|\mathbb{E}_{\theta^*} [\nabla_{\psi\lambda} \log p_{\theta^*}(Y)]\| \right. \\ & \quad \left. + \frac{8LB}{b} \frac{2L(1 + 8B/b)}{b^2} \|\hat{\theta}_1 - \theta^*\| \right\} \\ & \leq \|\hat{\theta}_1 - \theta^*\| \left\{ \frac{8LB}{b^2} + \frac{2L^2(1 + 8B/b)}{b^2} + \frac{8LB}{b} \frac{2L(1 + 8B/b)}{b^2} a_2 \right\}. \end{aligned}$$

The proof is then completed.  $\square$

## References

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